MODELO ARX PRUEBA ESCALÓN

IDENTIFICACION DE PROCESOS USANDO MINIMOS CUADRADOS

```
clear all
close all

Datos=xlsread('Datos_Escalon - Copy.xlsx');

t=Datos(:,1);
U=Datos(:,2);
Y=Datos(:,4);

subplot(2,1,1)
plot(t,U,'* k','MarkerSize',4)
ylabel('Entrada')
grid on
subplot(2,1,2)
plot(t,Y,'o b','MarkerSize',4)
ylabel('Salida')
xlabel('Tiempo')
hold on
grid on

% Seleccionando na=5, nb=5, d=1
s=length(Y);

X=[-[0;Y(1:s-1)],-[0;0;Y(1:s-2)],-[0;0;0;Y(1:s-3)],[0;U(1:s-1)],[0;0;U(1:s-2)],[0;0;0;U(1:s-3)
%X=[-[0;Y(1:s-1)],-[0;0;Y(1:s-2)],-[0;0;0;Y(1:s-3)],...
%    -[0;0;0;0;Y(1:s-4)],-[0;0;0;0;0;Y(1:s-5)],...
%    [0;U(1:s-1)],[0;0;U(1:s-2)],[0;0;0;U(1:s-3)],...
%    [0;0;0;0;U(1:s-4)],[0;0;0;0;0;U(1:s-5)],[0;0;0;0;0;0;U(1:s-6)]];

%Matriz de covarianzas
Q=X'*X
```

```
Q = 7×7
10^4 ×
    2.9678    2.8089    2.6401   -0.0738   -0.0738   -0.0719   -0.0688
    2.8089    2.8257    2.6194   -0.0701   -0.0701   -0.0701   -0.0682
    2.6401    2.6194    2.5730   -0.0650   -0.0650   -0.0650   -0.0650
   -0.0738   -0.0701   -0.0650    0.0020    0.0019    0.0018    0.0017
   -0.0738   -0.0701   -0.0650    0.0019    0.0019    0.0018    0.0017
   -0.0719   -0.0701   -0.0650    0.0018    0.0018    0.0018    0.0017
   -0.0688   -0.0682   -0.0650    0.0017    0.0017    0.0017    0.0017
```

```
%Vector de parámetros
Theta=inv(Q)*X'*Y
```

```
Theta = 7×1
    0.3205
```

```
    0.3732
   -0.1128
   18.8496
   18.6068
   14.2038
   12.0637
```

```
z=tf('z',0.1);
Gpm=((Theta(4)*z^3+Theta(5)*z^2+Theta(6)*z+Theta(7))/(z^3+Theta(1)*z^2+Theta(2)*z+Theta(3)))*(z
```

```
Gpm =

   18.85 z^3 + 18.61 z^2 + 14.2 z + 12.06
  ----------------------------------------
  z^4 + 0.3205 z^3 + 0.3732 z^2 - 0.1128 z

Sample time: 0.1 seconds
Discrete-time transfer function.
```

```
%Gpm=((Theta(6)*z^5+Theta(7)*z^4+Theta(8)*z^3+...
%    Theta(9)*z^2+Theta(10)*z+Theta(11)) / ...
%    (z^5+Theta(1)*z^4+Theta(2)*z^3+Theta(3)*z^2+...
%    Theta(4)*z+Theta(5)))...
%    *(z^(-1))
step(Gpm,'r--',2)
grid on
%Ym=step(Gpm,14);
Ym = lsim(Gpm,U,t)
```

```
Ym = 21×1
         0
   18.8496
   31.4159
   34.5575
   43.0500
   40.5732
   38.5515
   41.0815
   40.7460
   39.6812
      ⋮
```

```
%Validación del modelo
Sy=sum((Y-mean(Y)).^2)
```

```
Sy = 2.4965e+03
```

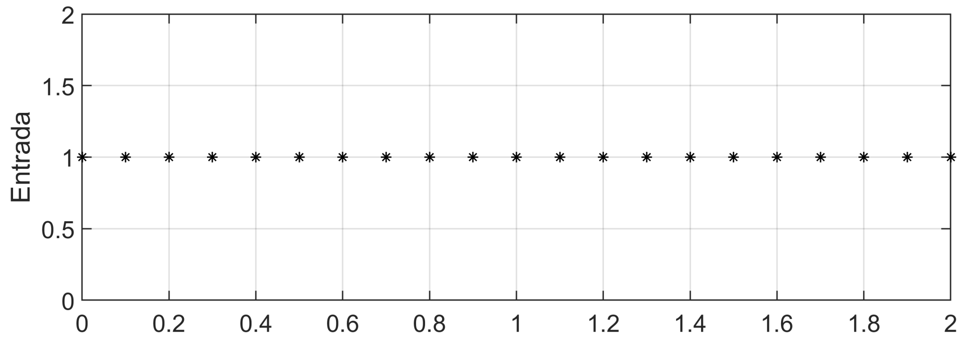
```
Sr=sum((Y-Ym).^2)
```

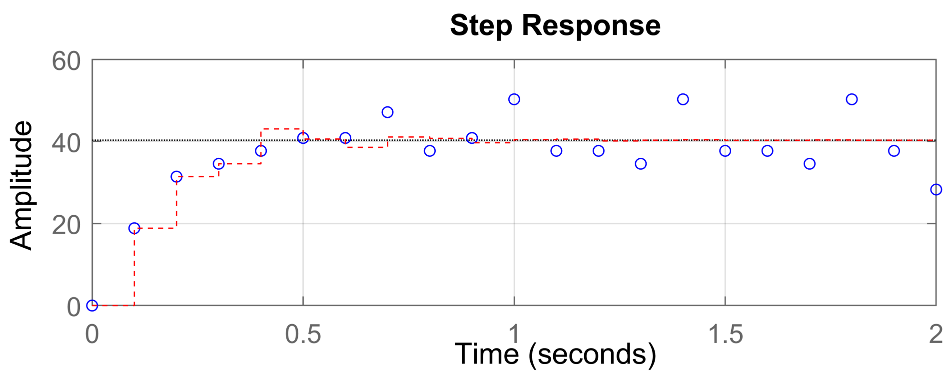
```
Sr = 619.6076
```

```
R2=(Sy-Sr)/Sy
```

```
R2 = 0.7518
```

```
hold off
```





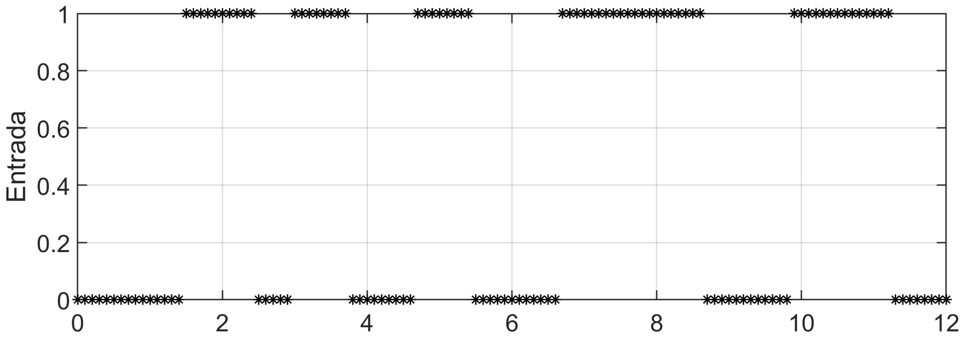
MODELO ARX PRUEBA PRBS

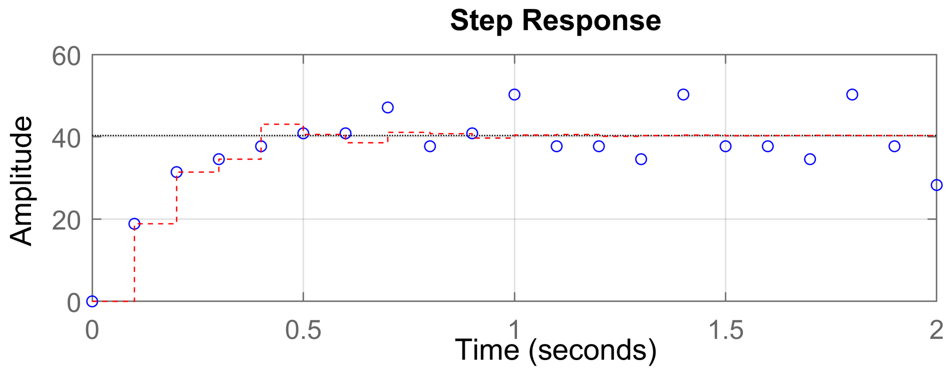
IDENTIFICACION DE PROCESOS USANDO MINIMOS CUADRADOS

```
Datos_prbs=xlsread('Datos_PRBS.xlsx');

t_prbs=Datos_prbs(:,1);
U_prbs=Datos_prbs(:,2);
Y_prbs=Datos_prbs(:,4);

subplot(2,1,1)
plot(t_prbs,U_prbs,'* k','MarkerSize',4)
ylabel('Entrada')
grid on
```






```
subplot(2,1,2)
plot(t_prbs,Y_prbs,'o b','MarkerSize',4)
ylabel('Salida')
xlabel('Tiempo')
hold on
grid on

% Seleccionando na=3, nb=3, d=1
s_prbs=length(Y_prbs);

X_prbs=[-[0;Y_prbs(1:s_prbs-1)],-[0;0;Y_prbs(1:s_prbs-2)],-[0;0;0;Y_prbs(1:s_prbs-3)],[0;U_prbs

%Matriz de covarianzas
Q_prbs=X_prbs'*X_prbs
```

```
Q_prbs = 7×7
10^4 ×
    9.5025    9.0327    8.4148   -0.2275   -0.2315   -0.2212   -0.2026
    9.0327    9.5025    9.0327   -0.2089   -0.2275   -0.2315   -0.2212
    8.4148    9.0327    9.5025   -0.1891   -0.2089   -0.2275   -0.2315
   -0.2275   -0.2089   -0.1891    0.0060    0.0055    0.0050    0.0045
   -0.2315   -0.2275   -0.2089    0.0055    0.0060    0.0055    0.0050
   -0.2212   -0.2315   -0.2275    0.0050    0.0055    0.0060    0.0055
   -0.2026   -0.2212   -0.2315    0.0045    0.0050    0.0055    0.0060
```

```
%Vector de parámetros
Theta_prbs=inv(Q_prbs)*X_prbs'*Y_prbs
```

```
Theta_prbs = 7×1
   -0.1364
    0.0122
   -0.0156
   27.8529
    6.3541
    0.7226
   -1.2351
```

```
z=tf('z',0.1);

Gpm_prbs=((Theta_prbs(4)*z^3+Theta_prbs(5)*z^2+Theta_prbs(6)*z+Theta_prbs(7))/(z^3+Theta_prbs(1
```

```
Gpm_prbs =

   27.85 z^3 + 6.354 z^2 + 0.7226 z - 1.235
  ------------------------------------------
  z^4 - 0.1364 z^3 + 0.01225 z^2 - 0.01557 z

Sample time: 0.1 seconds
Discrete-time transfer function.
```

```
%step(Gpm,'r--',12)
grid on
Ym_prbs=lsim(Gpm_prbs, U_prbs, t_prbs);
plot(t_prbs,Ym_prbs)
%Ym=step(Gpm,12);

%Validación del modelo
Sy_prbs=sum((Y_prbs-mean(Y_prbs)).^2)
```

```
Sy_prbs = 4.4119e+04
```

```
Sr_prbs=sum((Y_prbs-Ym_prbs).^2)
```

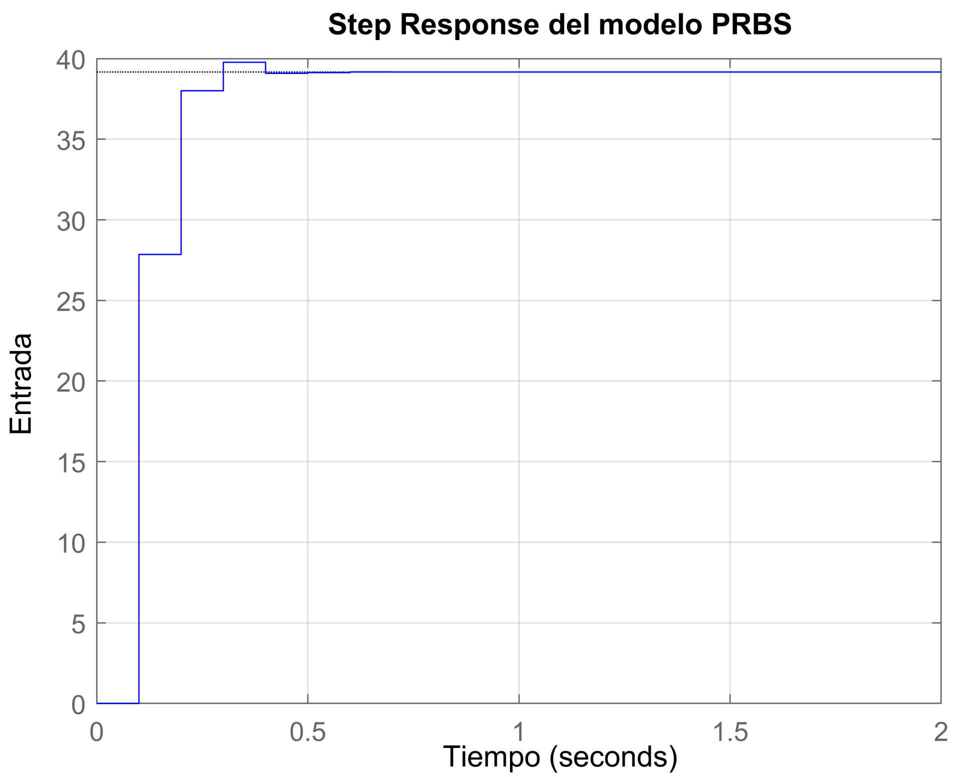
```
Sr_prbs = 4.8740e+03
```

```
R2_prbs=(Sy_prbs-Sr_prbs)/Sy_prbs
```

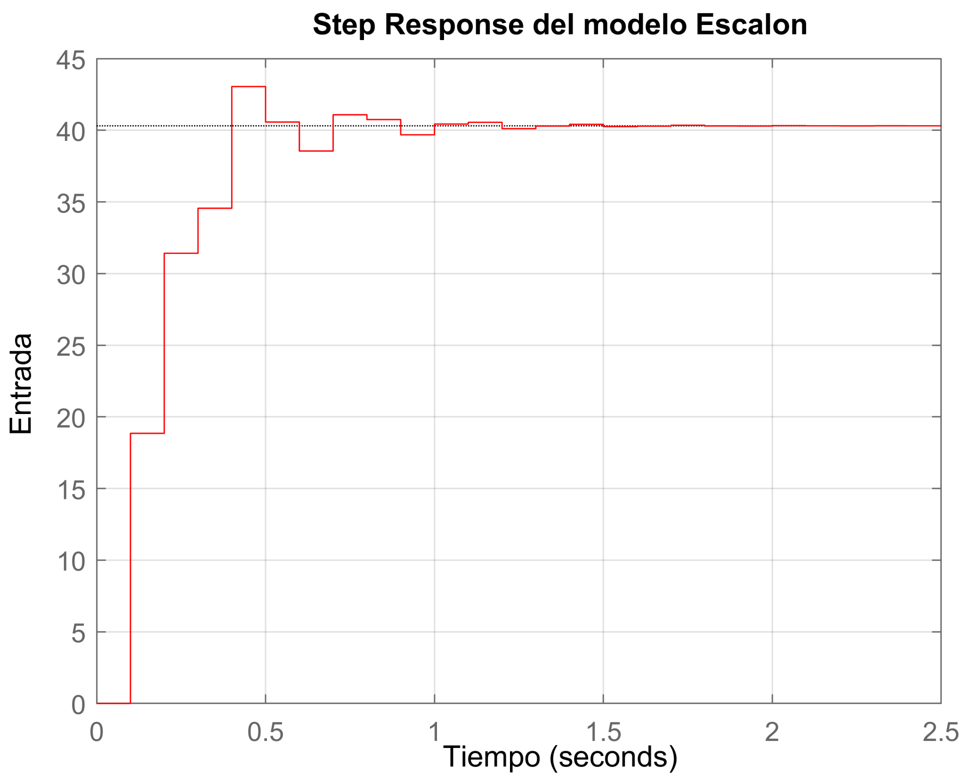
```
R2_prbs = 0.8895
```

```
hold off
close all
```

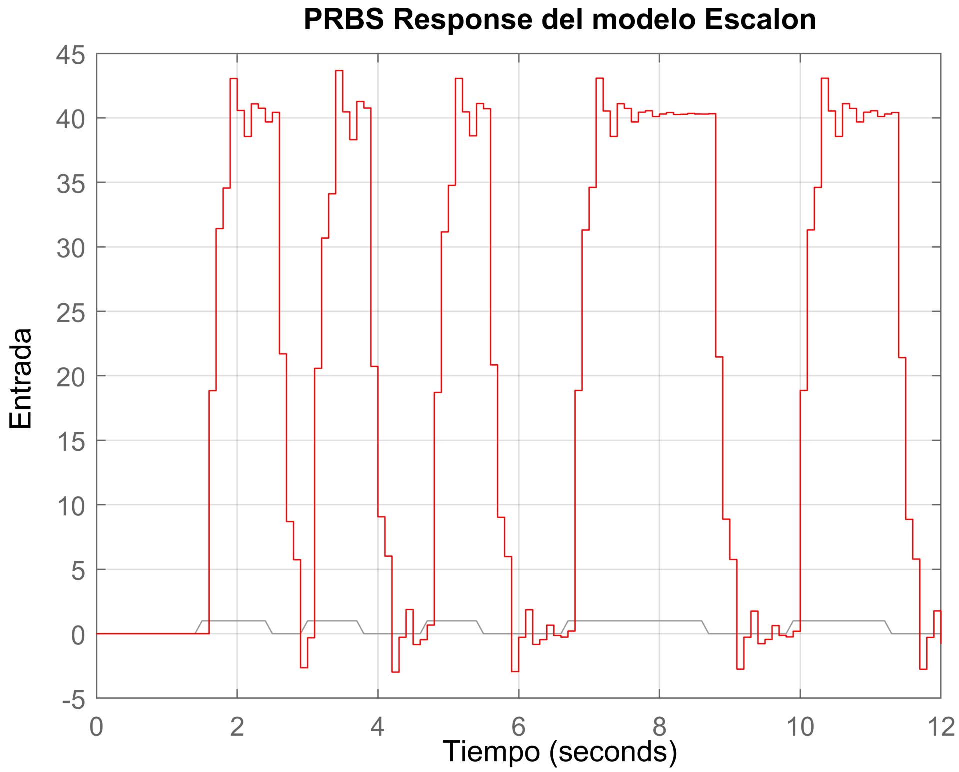
```
%Cruzados
step(Gpm_prbs,'b-')
title('Step Response del modelo PRBS')
ylabel('Entrada')
xlabel('Tiempo')
grid on
```



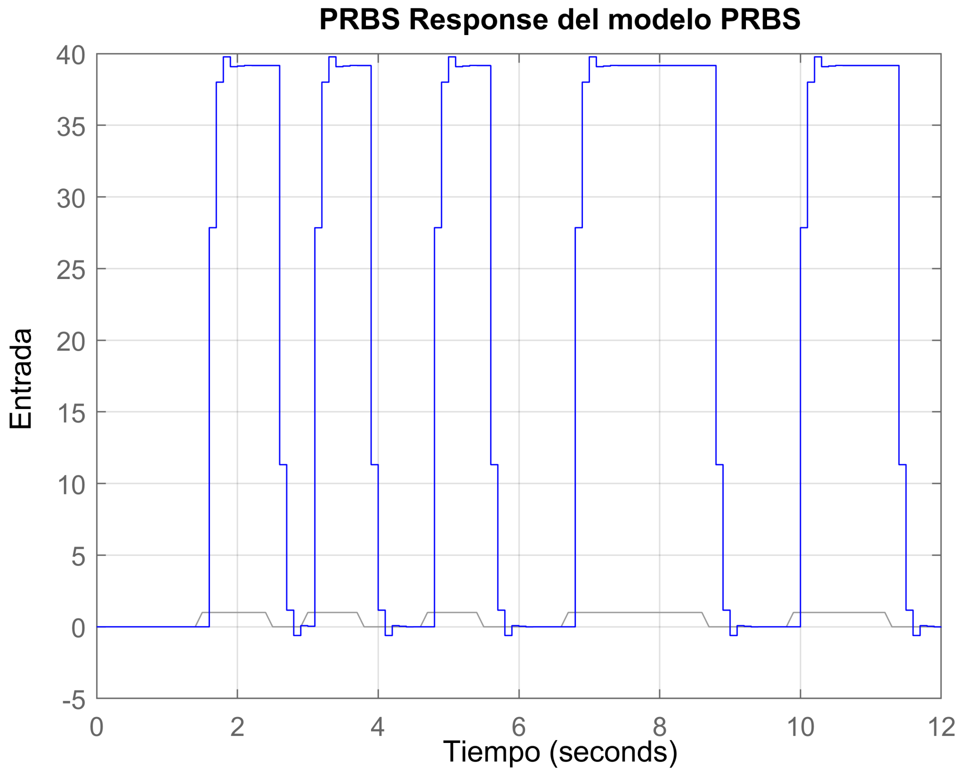

```
step(Gpm,'r-')
title('Step Response del modelo Escalon')
ylabel('Entrada')
xlabel('Tiempo')
grid on
```




```
lsim(Gpm,U_prbs,t_prbs,'r-')
title('PRBS Response del modelo Escalon')
ylabel('Entrada')
xlabel('Tiempo')
grid on
```




```
lsim(Gpm_prbs,U_prbs,t_prbs,'b-')
title('PRBS Response del modelo PRBS')
ylabel('Entrada')
xlabel('Tiempo')
grid on
```


PRBS Response del modelo PRBS
Entrada
Tiempo (seconds)
40
35
30
25
20
15
10
5
0
-5
0
2
4
6
8
10
12